



Mandate

An AI agent that can prove it's allowed to spend,
without saying who it works for.

01 Agents are starting to spend money

To trust an agent today, a seller has to know whose card or account is behind it. Every agent purchase becomes a record of who is buying what.

1

Who is behind it

The company or person the agent works for, on every purchase.

2

What they're doing

Five executives booked to one city says something. So does an agency's client list.

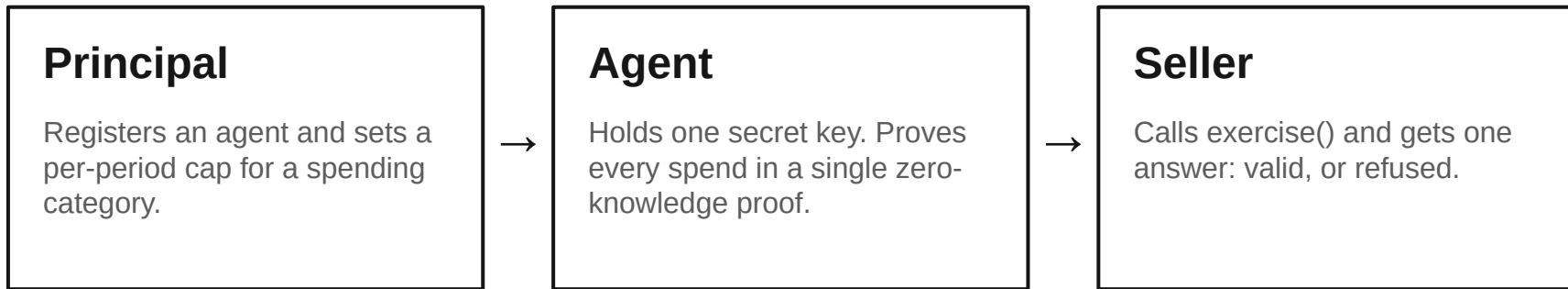
3

A permanent log

Every agent purchase becomes a trail tied to a real identity.

Visa, Mastercard and Google's AP2 all let owners cap an agent's spending. None of them hide who the owner is from the seller.

What Mandate does



In one proof, the agent shows:

- It is currently authorized by the real principal, for this category, and not revoked
- This request has not been used before (a nullifier stops replays)
- This period's spend plus this amount stays within the cap

If any check fails, the proof can't be built. Nothing is submitted and nothing is paid.

03 A correct refusal is the product

The real run on Midnight Preprod. Book a \$250 flight, then try a \$251 hotel on a \$500 budget. Fire the agent, then try a \$1 coffee.

5 Travel Booking Agent tries to book a \$251 hotel as agent Must fail Refused as required

\$250 + \$251 is one dollar over the \$500 cap. The budget is a running sum across bookings, not a per-booking check, and it's enforced to the dollar. Refusal here is the feature, not a bug.

Over budget: \$250 already spent, plus \$251 is more than the \$500 cap.
Refused on the agent's own machine. No valid proof exists, so nothing was submitted or paid.
failed assert: Mandate: category budget exceeded for this period

6 Fire Travel Booking Agent as principal Done

The kill switch. Existing grants aren't deleted from the tree — authorization is re-checked live on every single booking attempt instead, so revoking takes effect immediately.

7 Fired Travel Booking Agent tries to book a \$1 coffee anyway as agent Must fail Refused as required

The grant is still sitting in the Merkle tree and there's still budget left. It's refused anyway, purely because registration is checked live, every time — being in the tree once is never enough on its own.

Not authorized: Travel Booking Agent has been fired, so no booking can go through.
Refused on the agent's own machine. No valid proof exists, so nothing was submitted or paid.
failed assert: Mandate: agent is not currently authorized

What Midnight sees, and what it never sees

On the public ledger

- The agent's pseudonym (a hash of its key), on every spend
- Each category's cap, as a plain number
- Running totals and the amount of each spend
- Counts: grants issued, times exercised, active agents

Never on the ledger

- ☐ ~~Who the principal is (only a hash is stored)~~
- ☐ ~~The agent's real name~~
- ☐ ~~What a category means ("travel")~~
- ☐ ~~Who the agent bought from~~

Honest version: the budget numbers are public. What Mandate hides is who they belong to and what they're for.

05 Lose the laptop, keep the setup

On chain, the agent is only a hash. The mapping back to "Travel Booking Agent, \$500" is encrypted in the browser and stored on Cascade, Lumera's decentralized storage.

- AES-256-GCM in the browser; key derived from a Keplr signature
- Restored and decrypted from the command line: two separate crypto implementations agree
- The contract works the same with or without it

88139

Cascade action, recorded run (Sep 30)

88119

Cascade action, earlier run (Sep 17)

15

backup-crypto tests

06 It runs, and you can check it

7 / 7

walkthrough steps behave
as designed on Preprod

40

automated tests passing
(18 contract · 15 crypto · 7 UI)

1

"times exercised" after the run.
The 2 refusals never touched the chain

 **Mandate**
Spending limits for AI agents, enforced by zero-knowledge proof.

preprod ▾ Proving: wallet  lace mn_addr_sknfae3 [+ ↻]

01 **Your account** Refresh

DUST · FEE RESOURCE 15,000.00 1DUST available · cap 15,000 Generated continuously by your NIGHT. Every transaction spends a little.	NIGHT 3,000 1NIGHT Never spent on fees — it's the asset that generates DUST.	UNSHIELDED ADDRESS mn_addr_prep-psknfae3  Holds your NIGHT. This is where a faucet sends funds.	SHIELDED ADDRESS mn_shield-ad-zqkufh14  Your Zwap Identity. The SDK needs it for every contract call, but Mandate moves no tokens, so no shielded value is ever spent.
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02 **Contract** A Mandate contract on Preprod.

5c7f2cc3e8d8...0f969757  Explorer of Change contract

03 **Public ledger** Anyone can read these.

Contract on Midnight Preprod 49ef1c9bec0a...c19d55b9

Wallet-delegated proving through the DApp
Connector (Lace, 1AM). Keys never leave the
wallet.

Honest limits

The budget is public

Caps, running totals and amounts are plain numbers. Private budgets (commitments) are on the roadmap.

One agent's calls are linkable

The live revocation check discloses the agent's pseudonym every call. Not its real identity.

Circuit names show the purpose

registerAgent, exercise, revokeAgent... tell a reader this is agent authorization.

No real money moves yet

Mandate authorizes spending. Settlement through a payment rail comes later.

08 First market: travel booking agents

\$1.71T

business travel spend, 2026 (GBTA)

Why travel first

An agency's client list is its business. Its AI agent should be able to book for a client without the hotel or airline learning which company the client is.

30%

of travel bookings by AI agents by 2030 (IDC)

Same contract, other skins

- Company purchasing and procurement
- AI dev teams paying for APIs
- Ad agencies spending client budgets
- DAO and grant programs

Consumers are slower: about 2% of US consumers say they'd let an agent book fully on its own today (Skift).

Roadmap



Wave 2 · now

- Working app on Preprod
- Refusal-first UI and no-wallet demo
- Cascade lost-device backup
- Talk to independent travel agencies



Wave 3

- Unlinkable agents: revoke inside the Merkle tree, drop the public pseudonym check
- Private budgets: committed caps and totals
- A second skin: procurement or API spend



After

- One-line verifier SDK for booking engines and payment rails
- Pair with AP2 / x402 so the payment carries the proof
- Midnight Build Club, if selected



Check it yourself

Try it, no wallet

`tomiin.github.io/midnight-mandate`

Source (Apache-2.0)

`github.com/tomiin/midnight-mandate`

Contract, Preprod

`49ef1c9bec0a0e5e8ab6214184842ea15b6d68c50b6654d12e7ebedfc19d55b9`

Cascade backup

`action 88139 (earlier run: contract 5c7f2cc3..., action 88119)`